From theoretical to computational linguistics and back

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Captatio benevolentiae

LSA Summer School, Stanford, 1987



Vlieland, Spring 2024



Part I
From theoretical to computational linguistics

Machine translation - First generation

- string matching algorithm
- bilingual dictionary

(1) a. Reigers eten veel vis.
b. Herons eat much fish.

Problems

(2) a. Reigers eten nooit vis.
b. * Herons eat never fish
c. Herons never eat fish.

(3) a. Reigers eten veel kikkers.
b. * Herons eat much frogs
c. Herons eat many frogs.

(4) a. Reigers zingen niet.
b. * Herons sing not
c. Herons do not sing.

The solution: Addition of rules

- There is to be a difference of order in the translation from the order in the original, and an inversion is necessary.
- There is a problem of choice: the choice depends on an indication which follows the word under consideration.
- At a point where there is no equivalent word in the original text, a word is to be introduced into the translation.

Macdonald (1963)

These rules are added to the words in the bilingual dictionary.

The translation is more of a carefully prepared miracle than the result of the application of general rules.

Machine translation - Second generation

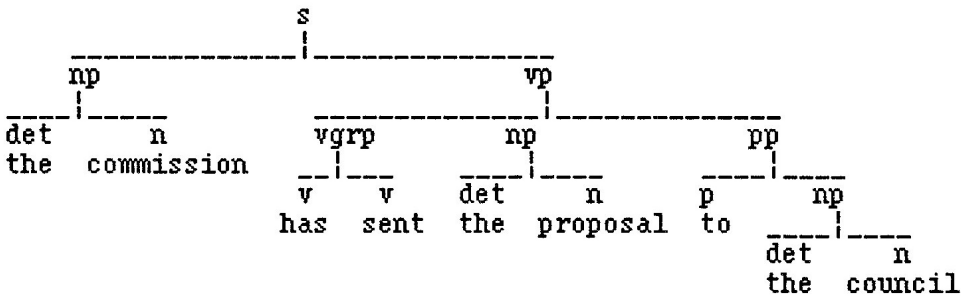
- The unit of translation is no longer the word but the syntactically analyzed sentence.
- Adjustments are not formulated in terms of strings but in terms of syntactic units.

EUROTRA

- an EU project on machine translation
- nine languages: Dutch, English, German, Danish, French, Italian, Spanish, Portuguese, Greek
- late seventies to early nineties

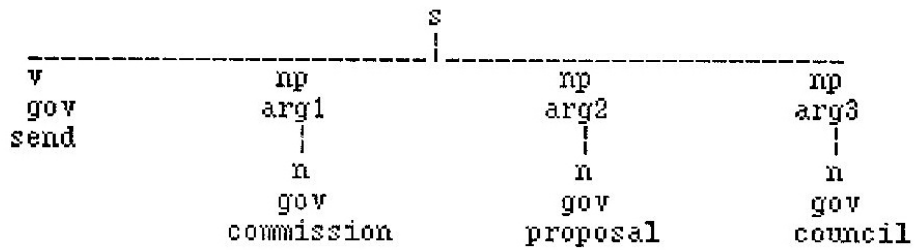
(5) The commission has sent the proposal to the Council.

CONSTITUENT STRUCTURE



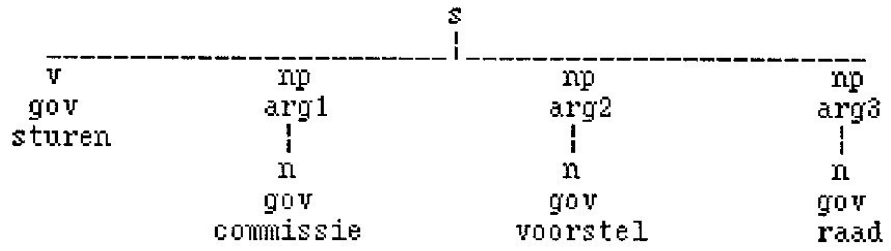
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INTERFACE STRUCTURE



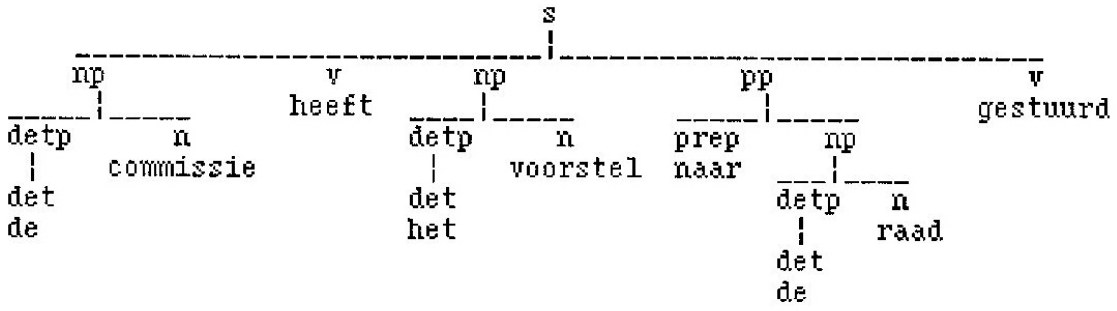
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INTERFACE STRUCTURE



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CONSTITUENT STRUCTURE



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Machine translation - Nowadays

- string matching algorithm
- bilingual corpus of source texts and their translation

No need for linguistic analysis

Theoretical linguistics no longer influences/inspires computational linguistics

Part 2
From computational to theoretical linguistics

Reference grammars for a broad audience

Quirk, Greenbaum, Leech & Svartvik, *A comprehensive grammar of the English language*. Longman, 1985.

Haeseryn, Romijn, Geerts, de Rooij & van den Toorn, *Algemene Nederlandse Spraakkunst*. Nijhoff & Wolters Plantyn, 1997.

Often with normative objectives

Reference grammars with a theoretical bias

Huddleston & Pullum, *The Cambridge grammar of the English language*. Cambridge University Press, 2002.

Broekhuis et al., *Syntax of Dutch*. Amsterdam University Press, 2012–2016.

- Nouns and Noun Phrases (2012) 2 volumes
- Adjectives and Adjective Phrases (2013) 1 volume
- Adpositions and Adpositional Phrases (2013) 1 volume
- Verbs and Verb Phrases (2016–18) 3 volumes

Common practice

- absence of normative objectives
- description of the system rather than of its use
- focus on one variant of the language
- introspection rather than observation
- little attention for lexical detail

Absence of normative aims

'Our aim is to **describe** and not **prescribe**: we outline and illustrate the principles that govern the construction of words and sentences in the present-day language without recommending or condemning particular usage choices.'

(Huddleston & Pullum, 2)

THE SYSTEM RATHER THAN ITS USE

'This book offers a description of the linguistic system itself.'

(Huddleston & Pullum, 2)

'Our goal of describing the internal structure of phrases and sentences means that we focus on competence (the internalized grammar of native speakers), and not on performance (the actual use of language)... This implies that ... we will not systematically incorporate the findings of currently flourishing corpus/usage-based approaches to language.'

(Broekhuis et al., viii)

The system rather than its use

Information about frequency of use is scarce and based on impressions of the authors.

'APs selected by a verb are normally predicative complements ... Non-predicative AP-complements may not exist at all, and, if they do, they are probably extremely rare.'

(Broekhuis et al., 329)

Focus on one variant of the language

'we will focus on the variety of Dutch that is spoken in the northwestern part of the Netherlands. One reason for doing this is that the authors who have contributed to the *Syntax of Dutch* are all native speakers of this variety.'

(Broekhuis et al., xii)

Introspection rather than observation

'This implies that we will make extensive use of constructed examples.'

(Broekhuis et al., viii)

'the authors ... can ... simply appeal to their own intuitions in order to establish whether this variety (of Dutch) does or does not exhibit a certain property.'

(Broekhuis et al., xii)

Introspection rather than observation

*	unacceptable
*?	relatively unacceptable compared to *
??	intermediate or unclear status
?	marked: not completely acceptable or disfavored form
(?)	slightly marked, but probably acceptable
%	varying judgments among speakers
#	unacceptable under intended reading
$	special status: old-fashioned, archaic, very formal, semantically incoherent, degraded/unacceptable for non-syntactic reasons, etc.

(Broekhuis et al., vi)

Little attention for lexical detail

Distinction between main verbs, auxiliary verbs and copular verbs. What are the copular verbs?

- *zijn* 'be', *worden* 'become' and *blijven* 'remain, stay'
- *lijken, schijnen* 'seem' and *blijken* 'appear'

(Broekhuis et al., Verbs and Verb Phrases, 248)

Even less than in normative grammars

- *zijn, worden* and a number of equivalents
- *blijven*
- *lijken, schijnen, blijken, heten, dunken*
- *staan* 'stand', *zitten* 'sit', *vallen* 'fall', ...
- *(ge)raken* 'get', *gaan* 'go', *komen* 'come', *lopen* 'run', ...

(Haeseryn et al., 1122)

A paradigm shift in descriptive linguistics

- more attention for matters of use
- more attention for language internal variation
- more reliance on observation than on introspection
- more attention for lexical detail

Treebanks

Syntactically annotated corpora (aka treebanks) are a byproduct of automated syntactic analysis.

In computational linguistics they are used to train and improve parsers.

In descriptive linguistics they can be used to provide empirical support for new style reference grammars.

Two kinds of treebanks

- High quality: (semi)-automatically produced, manually checked, externally validated
- High quantity: very large, automatically produced, not manually checked

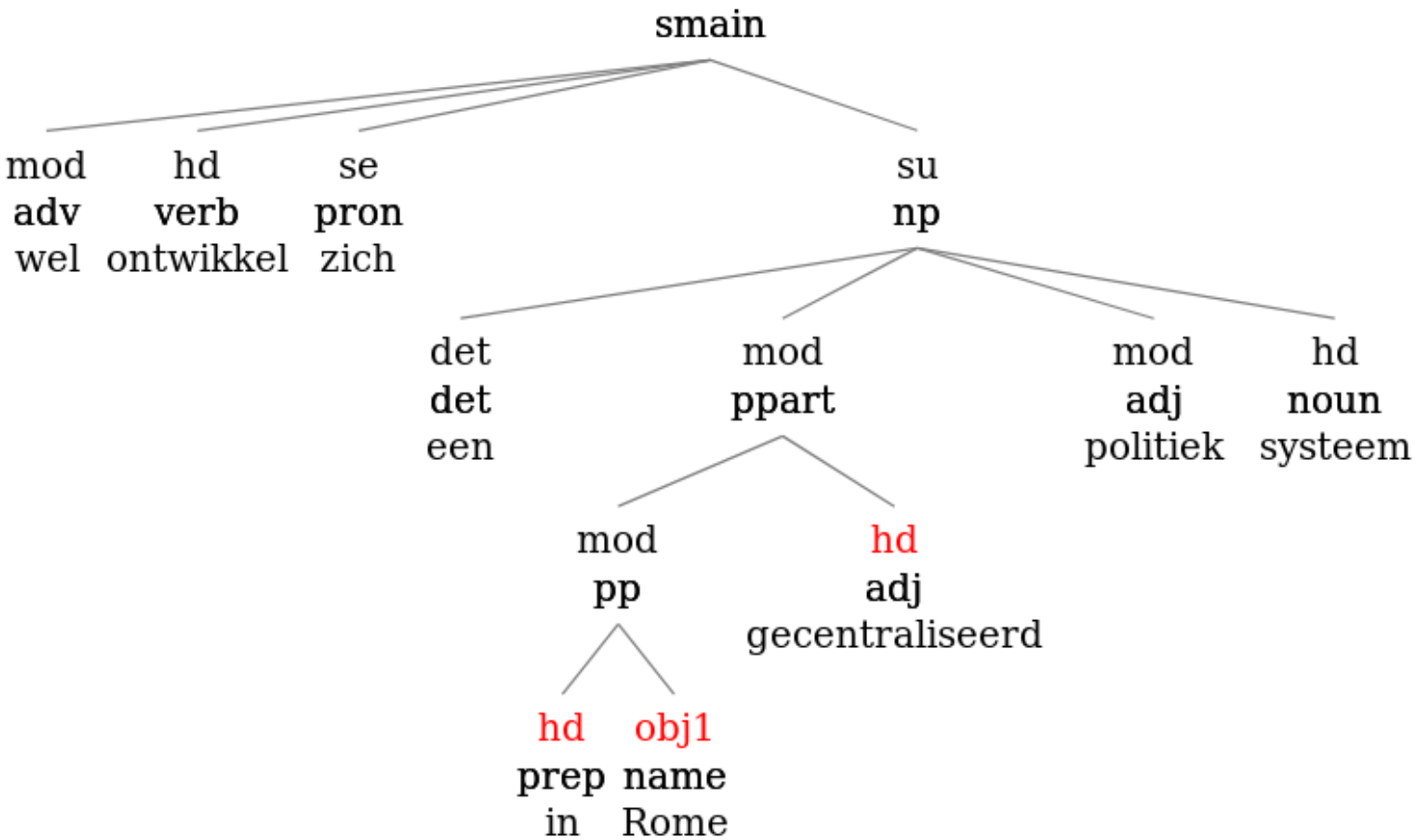
Two High Quality treebanks for Dutch

Spoken Dutch Corpus (appr. 1 M words)
Lassy Small (appr. 1 M words)

- 65,200 syntactically analyzed sentences
- each word is assigned a tag (98.63 % correct)
- each word is assigned a lemma
- each phrase is assigned a syntactic category
- each sign is assigned a dependency label (97.8 % correct)

ALPINO

(6) Wel ontwikkelde zich een in Rome gecentraliseerd systeem.
PTC developed REFL a in Rome centralized system
'A system developed that was centralized in Rome.'



A search tool : Gretel

XPath-queries

- //node[@lemma=”zijn”]
  24,676 hits
- //node[@lemma=”zijn” and @pt=”ww”]
  20,881 hits (84.62 %)
- //node[@cat=”np”]
  190,626 hits
- //node[@cat=”np” and @rel=”su”]
  39,393 hits (20.66 %)
- //node[@cat=”np” and node[@rel=”det”]]
  148,080 hits (77.68 %)
- //node[@cat=”np” and node[@rel=”det”] and node[@rel=”mod”]]
  76,265 hits (40.01 %)

Exploiting the treebank

What are the copular verbs?

Step 1: Identifying the predicative complements

- //node[@rel="predc"]

19,452 hits distributed over 17,186 sentences (26.36 % of the 65,200 sentences)

Step 2: Identifying the verbs that take a predicative complement

- //node[node[@rel="predc"] and node[@rel="hd" and @pt="ww"]]

19,153 hits (98.46 %)

Head hunting

Step 3: Collecting quantitative data about the verbs

- //node[node[@rel=”predc”] and node[@rel=”hd” and @pt=”ww” and @lemma=”zijn”]]
- //node[node[@rel=”predc”] and node[@rel=”hd” and @pt=”ww” and @lemma=”worden”]]
- //node[node[@rel=”predc”] and node[@rel=”hd” and @pt=”ww” and @lemma=”blijven”]]

Absolute frequency in LASSY Small

Prototypical		Equivalent	
zijn	12,914		
worden	1,475		
blijven	415		
		gaan	263
		(ge)raken	215
		staan	198
lijken	119		
		komen	105
blijken	102		
heten	93		
		zitten	72
		vallen	24
		lopen	11
schijnen	0		
dunken	0		

Relative frequency in LASSY Small

Copular use vs. other uses

- //node[@pt="ww" and @lemma="zijn"]
- //node[node[@rel="predc"] and node[@rel="hd" and @pt="ww" and @lemma="zijn"]]

20,881 hits for *zijn*
12,914 of those are combinations with a predicative complement

Relative frequency in Lassy Small

Prototypical	Copular	Total	%	Equivalent	Copular	Total	%
zijn	12,914	20,881	**61.85**				
worden	1,475	11,095	13.29				
blijven	415	961	43.18				
				gaan	263	2,315	11.36
				(ge)raken	215	326	**65.95**
				staan	198	1,108	17.87
lijken	119	354	33.62				
				komen	105	2,376	4.42
blijken	102	635	16.06				
heten	93	111	**83.78**				
				zitten	72	529	13.61
				vallen	24	527	4.55
				lopen	11	401	2.74
schijnen	0	22	0.00				

Food for thought

- The 15 verbs in the tables jointly account for 17,007 hits (88.8 % of the 19,153 combinations of a verb and a predicative complement)
- The treebank contains 250 other verbs with a predicative complement
- The distinction between prototypical copular verbs and their equivalents is not validated

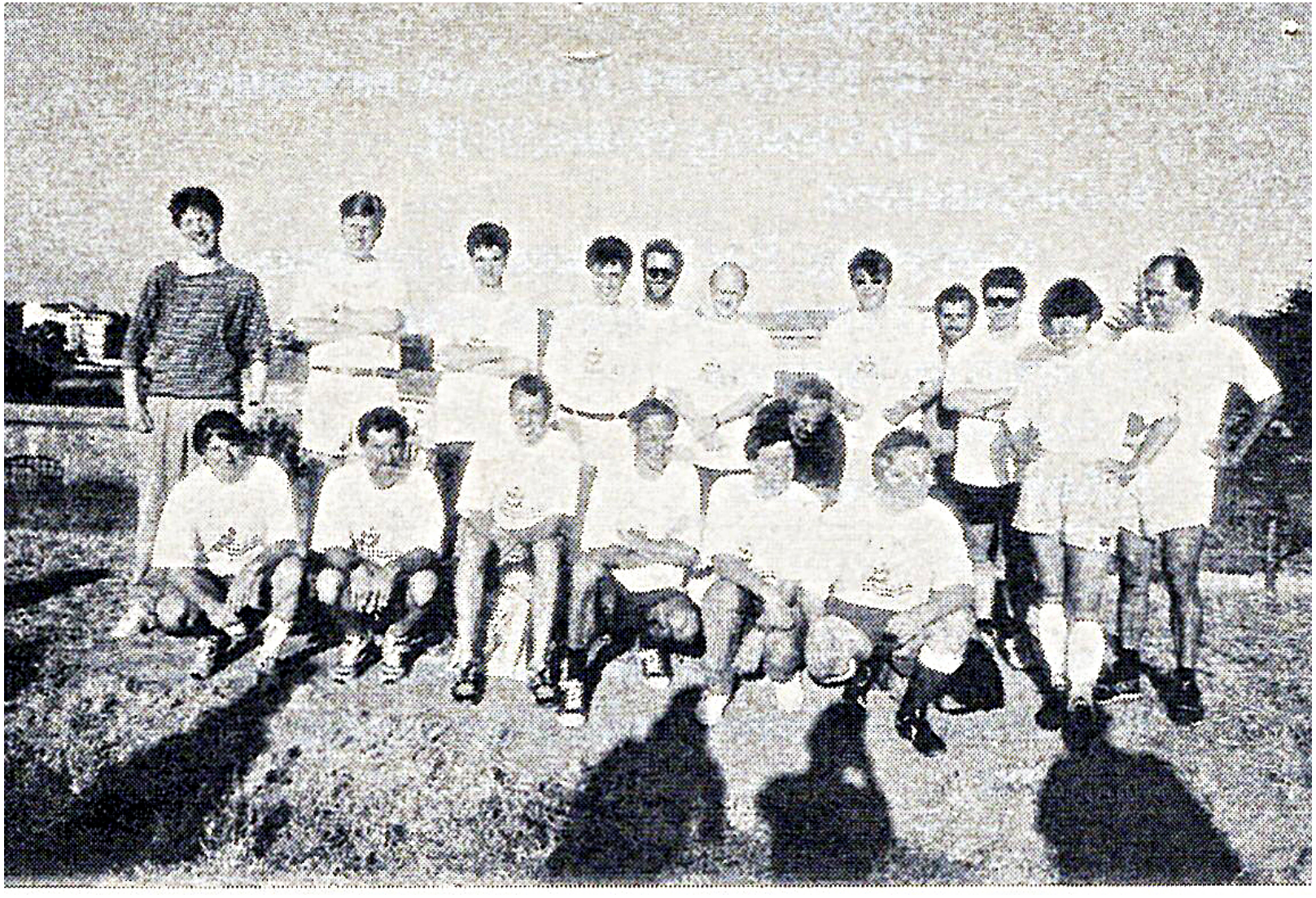
Wrapping up

- Treebanks can be used to identify the verbs with copular uses and to investigate their properties.
- The resulting picture is much more detailed than what is found in current descriptive grammars.

Conclusion

- The availability of large-scale high-quality treebanks,
- the availability of efficient tools to extract information from them, and
- the availability of linguists who know how to use the new resources
- has created the conditions in which a new style of descriptive and theoretical linguistics can flourish and thrive.

ESSLLI 1993, Lisbon



HPSG 2004, LEUVEN



Thank you (plural)